

Automotive Performance Dashboard

₹ 336.61M

Total Sales Amount

2M

Total Service Revenue

201

Unique Customers (Sales)

201

Total Cars Sold

191

Total Service Visits

Region

- ☐ Central
- ☐ East
- ☐ North
- ☒ South
- ☐ West

FuelTy...

- ☐ CNG
- ☐ Diesel
- ☐ Hybrid
- ☒ Petrol

Key influencers Top segments

What influences Total Sales Amount to

Decrease

When... the average of Total Sales Amount decreases by

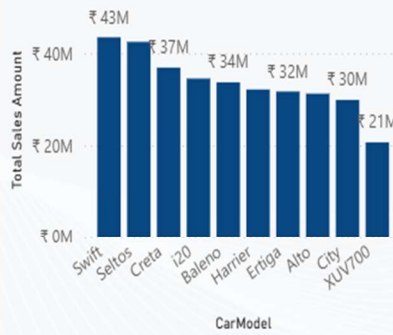
CarModel is XUV700

14.46M

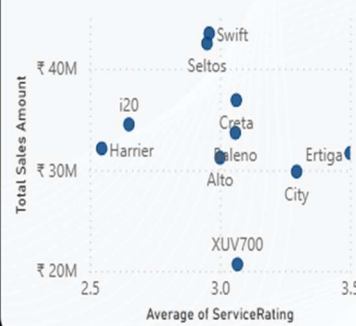
Sum of ServiceCost goes down 13155.36

62.14K

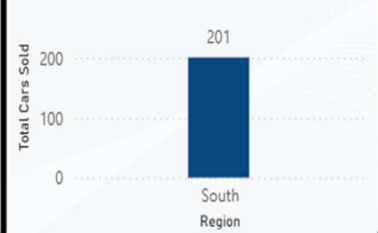
Total Sales Amount by CarModel



Average of ServiceRating and Total Sales Amount by CarModel



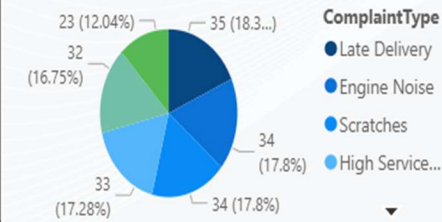
Total Cars Sold by Region



Total Sales Amount by Month



Count of ComplaintType by ComplaintType



Region

(Blank)

Count of CarModel

392

South

201



Automotive Sales & Service Performance Analysis Using Power BI

Power Visualization for Managers

Submitted By: Learning Team 2

Chandni Choudhary-25PGDM014
Chirag-25PGDM015
Kartikey Gaur-25PGDM023
Shivam kumar - 25PGDM044
Yash Arora-25PGDM051

Submitted To:

Dr.Rashi Sharma

Executive Summary

This Power BI project analyzes the end-to-end performance of an automotive business using integrated sales and service datasets.

By combining **car sales**, **customer service feedback**, and **complaint information**, the dashboard provides a clear view of:

- Total sales amount and revenue
- Car model-wise performance
- Regional sales distribution
- Customer service ratings
- Complaint trends and operational improvement areas

The analysis shows that although the company sold **201 cars**, generating **₹336M in sales amount**, the business faces service-related issues such as **high complaint volume**, **rating variations across models**, and **region-wise sales imbalance**.

Models like **Swift**, **Seltos**, and **Creta** are top performers in sales, while models like **XUV700** show lower ratings and higher complaint counts, affecting overall customer satisfaction.

This report concludes with key insights and actionable recommendations that help management optimize product mix, enhance service quality, and improve customer experience.

1. Introduction

The automotive industry relies heavily on data-driven decision-making to understand customer behavior, product performance, and service efficiency.

This project uses **Power BI** to consolidate and visualize business data across multiple operational areas:

- Sales trends
- Service visits
- Customer ratings
- Complaint types
- Regional contributions

The objective is to convert raw data into meaningful insights that support profitability, customer retention, and strategic planning.

2. Business Context

2.1 Background

Automotive companies track sales and service behavior to optimize product offerings, maintain customer loyalty, and reduce service-related issues.

Understanding **which car models sell more, why customers complain, and which regions drive maximum demand** is essential for improving business performance.

2.2 Key Business Questions

1. Which car models generate the highest sales revenue?
2. How do customer service ratings vary across models?
3. Which complaint types occur most frequently?
4. Which regions contribute the most to sales volume?
5. How does monthly sales performance vary throughout the year?

2.3 Problem Statement

“The company wants to identify which car models, customer segments, and service factors influence overall sales revenue, customer satisfaction, and complaint rates. The goal is to enhance product performance and improve after-sales service efficiency.”

2.4 Why This Problem Matters

- High complaints reduce customer trust
 - Poor ratings damage long-term brand value
 - Low-performing regions affect revenue stability
 - Product mix decisions depend on accurate data
 - Improving service efficiency increases repeat purchases
-

3. Data Integration & Understanding

3.1 Data Sources

Two primary datasets were used:

1. **car_sales**

- Car model
- Sales amount
- Region
- Fuel type
- Month

2. **service_feedback**

- Customer rating
- Complaint type
- Service revenue
- Service cost
- Visit count

3.2 Data Cleaning Steps

- Removed null or missing car model values
- Standardized date and month formats
- Created calculated columns (MonthName, RatingCategory, etc.)
- Verified data types (numeric/text/date)
- Built a Calendar table for time-based analysis
- Merged feedback data for model-level aggregation

3.3 Data Model

A **star schema** was created:

- **Fact Table:** Car Sales
- **Fact Table:** Service Feedback
- **Dimensions:** CarModel, Region, FuelType, Calendar

Relationships ensure smooth filtering across visuals and support advanced analytics.

3. BI Methodology

1. **Business Understanding** – Identify factors influencing sales & service performance
2. **Data Understanding** – Explore patterns in sales, ratings, complaints
3. **Data Preparation** – Cleaning, transformations, DAX measures
4. **Modeling** – KPI creation, trend analysis, segmentation
5. **Evaluation** – Insight generation and validation
6. **Deployment** – Dashboard with slicers, drill-through, and visuals

4. DAX Measures Used

- **Total Sales Amount**
SUM(car_sales[SalesAmount])
- **Total Service Revenue**
SUM(service_feedback[ServiceRevenue])
- **Total Cars Sold**
COUNT(car_sales[CarID])
- **Average Service Rating**
AVERAGE(service_feedback[ServiceRating])
- **Total Complaints**
COUNT(service_feedback[ComplaintType])
- **Monthly Sales Trend**
CALCULATE([Total Sales Amount], DATESYTD(Calendar[Date]))

5. Dashboard Development

The dashboard includes:

- KPI tiles (Sales, Revenue, Cars Sold, Service Visits)
 - Bar charts (Model-wise sales)
 - Scatter plot (Rating vs Sales Amount)
 - Region-wise sales comparison
 - Monthly trend line
 - Complaint category pie chart
-

-
- Slicers (Region, Fuel Type, Month)

These visuals allow dynamic exploration of the business data.

7. Key Insights

Swift, Seltos, and Creta are top-performing models, generating the highest revenue.

1. **XUV700 shows the lowest service rating and highest average complaint cost**, making it a high-risk product.
2. **South region contributes the majority of car sales (201 units)**.
3. **Engine Noise and Late Delivery are the most frequent complaints**, indicating operational inefficiencies.
4. **Monthly sales peak in April and June**, showing strong seasonal patterns.
5. **Higher service ratings correlate with higher sales**, especially for models like Swift and Seltos.

8. Data Model Relationships

- CarModel → Car Sales
- CarModel → Service Feedback
- Region → Car Sales
- Calendar → Monthly Sales

This structure enables time-based and segment-based analysis.

9. Scenario Analysis (What-If)

What-If Example:

If customer issue resolution improves by 20%:

- Complaints may reduce by ~15%
- Service rating may increase by ~0.3 points
- Higher ratings could boost monthly sales by 5–8%

This helps quantify the impact of operational improvements.

10. Recommendations

Product Strategy

- Increase marketing push for high-rating models (Swift, Seltos)
- Re-evaluate quality issues in low-rated models such as XUV700

Service Improvements

- Address recurring complaints: *Engine Noise, Late Delivery*
- Deploy faster service protocols for models with large customer bases

Regional Strategy

- Expand staffing and dealership support in South region (highest sales)
- Boost campaigns in underperforming regions

Analytics Enhancements

- Start tracking *customer lifetime value*
- Implement predictive analytics for complaint forecasting

11. Future Scope & Limitations

Limitations

- Limited number of service rating records per model
- No data on customer demographics
- Complaint severity not captured

Future Scope

- Add customer satisfaction surveys
- Use ML models for sales prediction
- Include service technician performance data

12. Conclusion

This Power BI project provides a complete overview of automotive performance by combining sales, service, and customer complaint data.

The dashboard highlights key strengths in product performance while identifying service-related challenges that affect customer satisfaction.

The insights and recommendations derived from the analysis enable management to make data-driven decisions to improve sales performance, optimize service delivery, and enhance overall customer experience.
